



State of the Services FY 2025 Q2 Report January 2025 - March 2025

Engagement Events Highlights

American Meteorological Society (AMS) Meeting New Orleans, LA | January 12-16

- NCEI participated in the AMS Annual Meeting, hosting two bilateral meetings to further discuss Artificial Intelligence (AI) for Public Good projects, as well as facilitating a panel on the state of AI in environmental sciences, weather forecasting, and climate monitoring.
- NCEI also contributed to presentations and engagements on NOAA's use of uncrewed systems and data management activities. In addition, they took the opportunity to support in person collaborations with NESDIS personnel regarding various issues related to Cloud Migration efforts.



Catastrophe (Cat) Risk Modeling Conference Orlando, FL | February 25-26

- NCEI hosted a NOAA Tools Cafe at the Reinsurance Association of America's (RAA) Cat Risk Modeling Conference. The Tools Cafe showcased the updated Storms Event Database, datasets on wind, storm footprint, and Billion Dollar Disasters (BDD).
- The meeting convened 700 participants, and industry members offered energetic support for the dedicated commitment NOAA has made to the sector, building on decades of NCEI's direct engagement with the RAA and their members. RAA members are experienced and well-established providers of reinsurance.



Product and Services Highlights



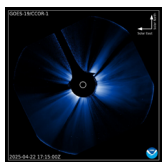
Hawaiian Data Included in Monthly Climate Reports

With the release of the January 2025 U.S. Climate Report, NCEI began summarizing the State of the Climate for Hawai'i. NCEI maintains monthly statewide, divisional, and gridded temperature and precipitation data for Hawaii from 1991 to 2025. This was made possible through a collaboration between NCEI and the University of Hawai'i's *Hawai'i Climate Data Portal*, filling a long-standing gap in NCEI's ability to report the State of the Climate across all 50 states.



Interactive StoryMaps

NCEI released two new StoryMaps—Hurricane Impacts and Billion Dollar Disasters (BDD)—on NOAA's website to show the impact of NCEI data on industry. The BDD map highlights the severe weather outbreak that devastated the Midwest in May 2024, with tornadoes, hail, and strong winds causing over \$20 billion in damages. The Hurricane Impact map explains hurricane formation, their effects on the U.S., and how NOAA data and tools help both the public and industry prepare for and respond to future storms.



NOAA Coronagraph Data Is Publicly Available

Compact Coronagraph-1 (CCOR-1) data is available through the NOAA Open Data Dissemination Platform. CCOR-1 is NOAA's first coronagraph, first operational instrument dedicated to space weather forecasting, first NCEI product to be fully operational in the NESDIS Common Cloud Framework, and the first operational coronagraph in the world. Hosted on the GOES-19 satellite platform, it is also the first instrument under the Space Weather Follow On Program to go into operation. CCOR-1 data is essential for monitoring coronal mass ejection speed and direction to determine whether they will hit the Earth and cause impacts to critical infrastructure, including the power grid.



Updated Tornado Climatology

The U.S. Tornado Climatology has been updated using data from NOAA's Storm Prediction Center, spanning the period from 1991-2020. Analysis of this 30-year dataset shows an annual average of 1,225 tornadoes nationwide, peaking in May with approximately 265 tornadoes and reaching a low in December with approximately 25 tornadoes.

Stakeholder Highlights



NCEI held a series of information sessions and demonstrations of the Atlas 15 pilot project to invite user feedback and facilitate co-development with the National Weather Service's Office of Water Prediction (OWP). User interest in this product is exceptionally high, and feedback has been robust and encouraging. Responses were documented by NCEI and provided directly to the Atlas 15 development team in the OWP. This activity is a proof of concept for the NOAA Service Delivery Model.



NCEI provided the overview of the local climate record for a gathering of the users of the Toolik Lake Long-term Ecological Research Station in Alaska's Arctic. The meeting was held in Santa Barbara, CA on the 50th anniversary of the station's founding and subsequent support by the National Science Foundation. Complexities of geography and discontinuous records in a data sparse region were discussed and users raised questions and expressed data needs.



The US-Japan Cooperative Program in Natural Resources Sea Bottom Surveys Panel held its 52nd meeting in Tokyo, Japan. The Sea Bottom Surveys Panel is a forum to promote the exchange of marine geophysical and bathymetric research and survey data, and related activities of marine science. NCEI gave reports on NCEI activities over the last year and an overview of the SoundCoop Passive Acoustic Monitoring Cyberinfrastructure Project.



NASA's Solar Dynamics Observatory held its annual meeting in Colorado and multiple NCEI staff were in attendance and presented information. Members of NCEI presented an overview of NOAA's Solar Observatories, including the One-NOAA Space Weather (SWx) Strategy, NOAA's SWx groups, NCEI's role therein, and how to access NOAA's SWx data. NCEI held a hands-on workshop for attendees to learn more about and work with GOES-R data. NCEI also presented a talk about GOES-R EXIS on-orbit calibrations at the Extreme Ultraviolet Calibration Workshop. Additionally, NCEI presented on the evolution of solar coronal dimmings and their relationship to coronal mass ejections, and how dimmings could be potentially used for space weather forecasting.

Upcoming Engagement

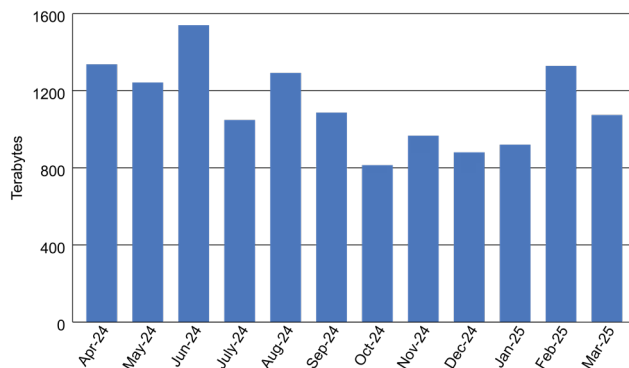
American Association of State Climatologists
Annual Meeting (Virtual)

Jun. 10-13

Customer Reach

NCEI Makes Data Accessible

- 1.2B web hits and 3.3k TB of data delivered online*



*Analytics include model & satellite data; data statistics may not be comprehensive across NCEI (e.g. CLASS)

U.S. Sectors Served

- Top Sectors



Unique Customer Requests

Southern Border Data Inquiry:

NCEI responded to an inquiry from the North American Aerospace Defense Command for information that will assist the Command in planning for anticipated seasonal impacts on troops deployed to the Southern Border. NCEI data resources such as temperature and extreme precipitation were shared.

Data Request for Potomac River Tragedy:

A CNN Reporter requested data to portray the depth of the Potomac River near the site of the tragic plane and helicopter crash. NCEI responded with information on how to use NCEI's high-resolution Continuously Updated Digital Elevation Models to generate a map of the Potomac River and provided an example map.



Fans



136.5k



63.8k



25.3k